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ORIGINAL ARTICLE

Selective delipidation of *Mycobacterium bovis* BCG retains antitumor efficacy against non-muscle invasive bladder cancer

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Abstract

Purpose Repeated instillations of bacillus Calmette *et* Guérin (BCG) are the gold standard immunotherapeutic treatment for reducing recurrence for patients with high-grade papillary non-muscle invasive bladder cancer (NMIBC) and for eradicating bladder carcinoma-in situ. Unfortunately, some patients are unable to tolerate BCG due to treatment-associated toxicity and bladder removal is sometimes performed for BCG-intolerance. Prior studies suggest that selectively delipidated BCG (dBCG) improves tolerability of intrapulmonary delivery reducing tissue damage and increasing efficacy in preventing *Mycobacterium tuberculosis* infection in mice. To address the lack of treatment options for NMIBC with BCG-intolerance, we examined if selective delipidation would compromise BCG's antitumor efficacy and at the same time increase tolerability to the treatment.

Materials and methods Murine syngeneic MB49 bladder cancer models and in vitro human innate effector cell cytotoxicity assays were used to evaluate efficacy and immune impact of selective delipidation in Tokyo and TICE BCG strains.

Results Both dBCG-Tokyo and dBCG-TICE effectively treated subcutaneous MB49 tumors in mice and enhanced tumor-infiltrating $CD8^+$ T and natural killer cells, similar to conventional BCG. However, when compared to conventional BCG, only dBCG-Tokyo retained a significant effect on intratumoral tumor-specific $CD8^+$ and γδ T cells by increasing their frequencies in tumor tissue and their production of antitumoral function-related cytokines, i.e., IFN-γ and granzyme B. Further, dBCG-Tokyo but not dBCG-TICE enhanced the function and cytotoxicity of innate effector cells against human bladder cancer T24 in vitro.

Conclusions These data support clinical investigation of dBCG-Tokyo as a treatment for patients with BCG-intolerant NMIBC.

Keywords BCG · Delipidation · Bladder cancer · Treatment

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Introduction

Intravesical attenuated *Mycobacterium bovis* bacillus Calmette *et* Guérin (BCG) is indicated for patients with intermediate and high-risk non-muscle invasive bladder cancer (NMIBC) [1]. Initiated soon after transurethral biopsy/resection, complete BCG induction and maintenance includes 27 intravesical instillations of live bacteria over a period of 3 years per patient. Level 1 clinical evidence shows that BCG maintenance prevents disease relapse and also reduces the risk of disease progression [2]. However, many patients are unable to tolerate repeated BCG instillations due to treatment-associated side effects, such as urinary frequency, urinary urgency, bladder pain, and bladder spasms [3, 4]. Patients with BCG-intolerance have limited treatment options and often undergo cystectomy [5, 6].

To address BCG intolerance, this study examines a modified formulation of BCG designed to decrease BCG's toxicity without compromising its efficacy. Selectively delipidated BCG (dBCG) by organic solvent petroleum ether (PE) was developed to be a less immunopathogenic vaccine administered directly into the lungs, where it attenuates pulmonary inflammation and increases protection against *Mycobacterium tuberculosis* infection in mice significantly reducing associated tissue damage [7]. PE-delipidation on BCG cell envelope only targets potent immune-modulating lipids such as trehalose dimycolate (TDM) [8], di- and tri-acylglycerols (DAGs/TAGs) [9], phthiocerol dimycocerosates (PDIMs) [10], and phenolic glycolipids (PGLs) [11]. These apolar lipids can induce rapid and strong inflammatory responses, causing immunopathology [7, 8], which could be highly relevant in BCG's toxicity in the bladder. Virulent *Mycobacterium* spp. also utilize these lipids to undermine host immunity [12–14]; thus, we hypothesized that removal of apolar lipids is expected to result in reducing undesired adverse effects in the bladder, while favoring local anti-tumor immune responses. To evaluate the toxicity and antitumor efficacy of dBCG compared to conventional BCG, this study used a murine model of bladder cancer and in vitro human bladder cancer cell cytotoxicity assays.

Materials and methods

Ethical statement

All animal experimental procedures were approved by the UT Health San Antonio Institutional Animal Care and Use Committee (IACUC) (#20120040AR). Collection and use of clinical samples from bladder cancer patients were approved and abided by institutional review boards (IRB) protocol HSC2012-159H following the Helsinki Declaration principle. Written consent forms were obtained.

Animals

C57BL/6 wild type (WT) female mice (The Jackson Laboratories, Bar Harbor, ME) aged 6 to 8 weeks were housed under conventional condition. Mice in each study were randomized into three groups on day 5 post-tumor challenge: phosphate-buffered saline (PBS)/control, PBS-treated-BCG (BCG), or PE-treated-BCG (dBCG). Each group contained 4–5 mice ($n = 8$–10 tumors) in each experiment, which was repeated 2–3 times independently. Power analysis was used to estimate the minimum sample size required for each or pooled experiments.

Live bacteria stock preparation and selective delipidation of *Mycobacterium bovis* BCG strains

Mycobacterium bovis BCG-Tokyo (Japan BCG Laboratory, Tokyo, Japan) and *M. bovis* BCG-TICE strains (Merck & Co, NJ) were grown and selectively delipidated as described [7]. Briefly, fresh BCG grown on 7H11 (Remel) agar supplemented with oleic acid, albumin, dextrose, and catalase (OADC) enrichment plates for up to 2 weeks was harvested, suspended in 1 ml of PE (Fisher Chemical) or PBS as control with 2 min vortex at maximum speed, followed by incubation time of 5 min at room temperature (RT) and centrifugation at 6,000 g for 5 min. These steps were repeated 3 times. The supernatant containing PE-extracted lipids were dried with a nitrogen evaporator (Organomation) and stored at − 20 °C until further analysis for lipid content. PE- or PBS-treated BCG pellets were evaporated briefly in the biosafety cabinet to remove residual solvent, re-suspended in 7H9 (BD-Difco) broth media containing 20% sterile glycerol (Sigma-Aldrich) and stored at − 80 °C in aliquots prior to use. Viability of BCG/dBCG from representative frozen stock vials was determined by counting colony-forming unit (CFU) after thaw and culture in 7H11 plus OADC supplemented agar plates at 37 °C for approximately 3 weeks. On each treatment day, BCG control and dBCG stock vials were thawed on ice and centrifuged at 10,000 g, 4 °C for 10 min. Bacterial pellets were resuspended in sterile endotoxin free PBS at desired concentration.

Analysis of lipid extractions from BCG cell envelope

As described [7], after sequential PE extraction, dBCG was further extracted with chloroform/methanol (C/M) (2:1, v/v) at 37 °C for 12 h. PE and C/M extracted lipids were analyzed by thin layer chromatography (TLC) using aluminum-backed TLC plates with solvent systems targeting different lipid compounds: C/M (95:5, v/v) for TDM, mycoside B (MycB), PGL; PE: acetone (96:4, v/v) for PDIMs and TAGs; chloroform/acetic acid/methanol/H_2O (40:25:3:6, v/v/v/v) for phosphatidyl-*myo*-inositol mannosides (PIMs). Lipids were visualized as TLC bands with 10% H_2SO_4 in ethanol spraying followed by charring at 100 °C. Developed TLCs with lipid bands were imaged by an Epson Perfection V600 photoscanner and quantified using the software ImageJ (National Institutes of Health, NIH).

Cell line and culture

All cell lines were tested and confirmed free of mycoplasma by Mycoalert PLUS Detection Kit (Lonza Bioscience). The mouse bladder cancer cell line MB49 was a gift from Dr. Jeffery Schlom (NIH). The human bladder cancer cell line T24 was a gift from Dr. Rita Ghosh at UT-Health San Antonio. Both cell lines were maintained in complete DMEM media (Thermo Fisher Scientific) containing 10% fetal bovine serum (FBS, HyClone/Thermo-Fisher Scientific), 100 IU/ml of penicillin (Corning), 100 mg/ml of streptomycin (Corning), and 2 mM of fresh L-glutamine (Corning) at 37 °C, in 5% CO_2 and cultured for at least 1 passage before use.

Mouse tumor model and BCG treatment

For subcutaneous (s.c.) model, MB49 cells were harvested and injected subcutaneously into mice at 2×10^5 cells in 100 µl PBS at both flanks on their lower back with hair shaved. Tumor volume was measured 3 times per week by a caliper and calculated using the formula: volume = length × width × width/2 (mm^3). Starting on day 5 post-MB49 tumor challenge, 1×10^6 CFU of BCG or dBCG in 50 µl of PBS were injected intratumorally into each s.c. tumor, weekly for 3 weeks. For orthotopic model, female mice were anesthetized inside isoflurane/O_2 chamber. Bladder from each mouse was emptied and instilled with 100 µl of poly-L-lysine (Sigma-Aldrich) through a urethral catheter (24G/0.56 inch, Becton Dickinson) for 30 min (50 µl of catheter volume was included). Then, poly-L-lysine was drained from the bladder with catheter still inserted, followed by instillation with 80,000 MB49 cells in 50 µl of PBS per bladder for 1 h (plus 50 µl of catheter volume). On day 1, 8, 15, 22, 29, BCG instillations were performed in similar procedure via urethral catheter at 3×10^6 CFU in 50 µl of PBS per bladder (plus 50 µl of catheter volume) for 2 h.

Mouse tumor processing for immune analysis by flow cytometry

Tumor tissue was harvested at the end of study and processed for staining as described [15] using reagents listed in supplemental Table S1. Briefly, tumor was weighed and crushed in 3 ml of serum-free RPMI media (Corning) per 0.5 g of tissue, with 1.5 mg/ml of collagenase IV (Sigma-Aldrich) and 0.25 mg/ml of DNase I (Sigma-Aldrich), respectively, for incubation at 37 °C, in 5% CO2 for 1 h, followed by addition of 7 ml of complete RPMI media (cR-10) containing 10% FBS, 2 mM of L-glutamine, penicillin (100 IU/ml) plus streptomycin (100 mg/ml) and passing through 100 µm strainer to obtain single cell suspensions (SCS) for live cell counting and cryopreserved in copies with freezing media (10% DMSO in complete RPMI media with 50% FBS) for flow cytometry analysis later. Samples in SCS at 1×10^6 cells per 100 µl of flow buffer (PBS with 2% FBS) were either stained directly for cell surface markers or stimulated first with Cell Activation Cocktail (BioLegend) in 100 µl of cR-10 at 37 °C for 5 h, followed by cell surface staining and then intracellular staining (supplemental Table S1). Fluorescence minus one (FMO) control was included for each marker, with gating threshold of $\leq 1\%$ for positive population. Samples were analyzed on LSRII cytometer (BD) using FACS Diva Software vr. 8.0.1 (BD) and FlowJo Software (vr.10). Absolute number (AN) of any population or subset per 0.1 g of tumor tissue was calculated by multiplying its percentage by AN of total live cells per 0.1 g of tumor tissue.

Mouse serum collection

Blood collection was performed by bleeding from submandibular vein (cheek pouch) using 5-mm animal lancet (Goldenrod, MEDIpoint) at each time point (baseline, post-2nd BCG and post-3rd BCG treatment). Blood was clogged at RT for at least 1 h, or at 4 °C overnight, then centrifuged at 10,000 g, 4 °C for 10 min. Supernatant/serum was carefully removed and placed in a new microcentrifuge tube and stored in aliquots at − 80 °C for further analysis.

Mouse tumor cultured supernatant collection

Fresh or thawed tumor SCS were resuspended in 150 µl of cR-10 normalized as per 4 mg of equivalent original tumor tissue and cultured at 37 °C for 18–24 h. Supernatants were collected by centrifugation at 10,000 g, 4 °C for 10 min and then, stored in aliquots at − 80 °C for further analysis.

Mouse cytokine/chemokine Luminex assay

Serum supernatant (diluted 2–4 folds) and/or undiluted supernatant from 24-h cultured tumor tissue SCS were used for mouse ProcartaPlex Luminex assay (Invitrogen/Thermo-Fisher Scientific). All samples were run and analyzed on FLEXMAP-3D (Luminex) with xPONENT software vr. 4.0

(Luminex), in the BASiC core at UT-Health San Antonio (RP150600).

Mouse CRP ELISA

Serum and supernatant from 24-h cultured tumor tissue SCS were diluted in 1000 or 250-folds, respectively, for mouse C-reactive protein (CRP) ELISA (Thermo Fisher Scientific) and analyzed using a Synergy 2 plate reader (BioTek) with Gen5 vr. 1.05 Software (BioTek).

Preparation of fresh human peripheral blood mononuclear cells (PBMCs)

Whole blood from patients enrolled in our bladder cancer repository (BCR) was collected and processed as described [16]. After plasma removal, fresh PBMCs were isolated by Ficoll-Paque (GE, Healthcare) gradient centrifugation and total viable cells were determined by an automatic cell counter (Vi-CellXR, Beckman Coulter) prior to innate effector expansion culture or cryopreservation.

Human innate immune cells expansion for in vitro cytotoxicity and immune analyses

Innate immune cells (mainly γδ T cells and natural killer/NK cells) were expanded from PBMCs of at least 3 bladder cancer patients [16]. Briefly, fresh PBMCs were cultured in cR-10, containing 15 µM of isopentenyl pyrophosphate (IPP, Sigma-Aldrich) and 100 IU/ml of IL-2 (BioLegend), with or without live BCG or dBCG (Multiplicity of Infection/MOI = 0.1). On day 14 of culture, effector cells were harvested and mixed with CFSE-labeled (Molecular Probes, Life Technologies) target cell T24 at effector to target (E:T) ratio of 25:1 with a final target cell number of 2×10^4 on 96-well U-bottom plates and incubated at 37 °C for 5 h. Mixed cells were stained with fixable viability dye (FVD)-eFluor455UV and analyzed on a BD LSRII flow cytometer using FACS Diva Software. Target tumor cell death (%) was based on the gating of FVD positive population over CFSE positive population. A portion of effector cells were also stained and analyzed for immune profile as described above (supplemental Table S1).

Statistical analysis

Tumor growth curves between groups were compared with two-way analysis of variance (ANOVA). To compare any two groups with only one variable, the D'Agostino & Pearson omnibus normality test was used; if $p > 0.05$, then a parametric t-test was used; if $p < 0.05$, then nonparametric Mann–Whitney test was used. If one group of data passed normality and the other did not, nonparametric test was used; if any group had $n \le 7$ (*N* too small), unless data sets had substantial outliers (non-parametric distribution), it was considered passed normality test. P values were two-sided, and $p < 0.05$ was considered statistically significant. Statistical analyses were performed with GraphPad Prism vr. 5 and 6.

Results

Conventional BCG and dBCG have comparable treatment efficacy against bladder cancer

To evaluate the treatment efficacy of dBCG compared to conventional BCG, C57BL/6 WT female mice were challenged subcutaneously with MB49 bladder cancer cells and monitored for tumor growth (Fig. 1a). Conventional Tokyo and TICE BCG strains significantly suppressed tumor growth compared to PBS control (Fig. 1b, c). Both dBCG-Tokyo and dBCG-TICE also suppressed MB49 tumor growth with no significant differences compared with their respective conventional BCG strain (Fig. 1b, c). Impact of selective delipidation on BCG's efficacy was also examined in orthotopic MB49 model (Fig. 1d). Although there was no significant difference between the effect of BCG-Tokyo and dBCG-Tokyo on the survival of WT mice bearing orthotopic MB49 tumors (supplemental Fig.S1A), mice treated with dBCG-Tokyo had significantly lower bladder weight than those treated with conventional BCG-Tokyo or PBS (Fig. 1e). These data suggest that selective delipidation does not compromise BCG efficacy in treating bladder cancer.

dBCG-Tokyo induces less inflammation compared with conventional BCG-Tokyo

To investigate the effect of BCG selective delipidation on systemic and local inflammation, mouse urine from orthotopic model, mouse serum and tumor supernatant from s.c. model were assessed for markers of inflammatory and adaptive immune responses. In mice bearing orthotopic bladder tumors, significant reduction in urine IL-6 to IL-10 ratio in dBCG-Tokyo group *vs.* BCG-Tokyo was only observed by d15/ third week of BCG instillation (Fig. 2a), when the ratios of urine IL-6 to IL-10 were the highest in both control and BCG-Tokyo groups (supplemental Fig.S1B). There was lower yet non-significant trend of the ratios of IL-1β to IL-10 by d15 and TNF-α to IL-10 by d1 in the urine of dBCG-Tokyo group compared with BCG-Tokyo group (Fig. 2b, c; supplemental Fig.S1B). Overall, in the serum compared to conventional BCG in s.c. tumor model, dBCG-Tokyo had no significant effect on serum cytokines involved in T helper cell differentiation and pro-inflammatory/inflammatory function (supplemental Fig.S2A), except for IL-10,

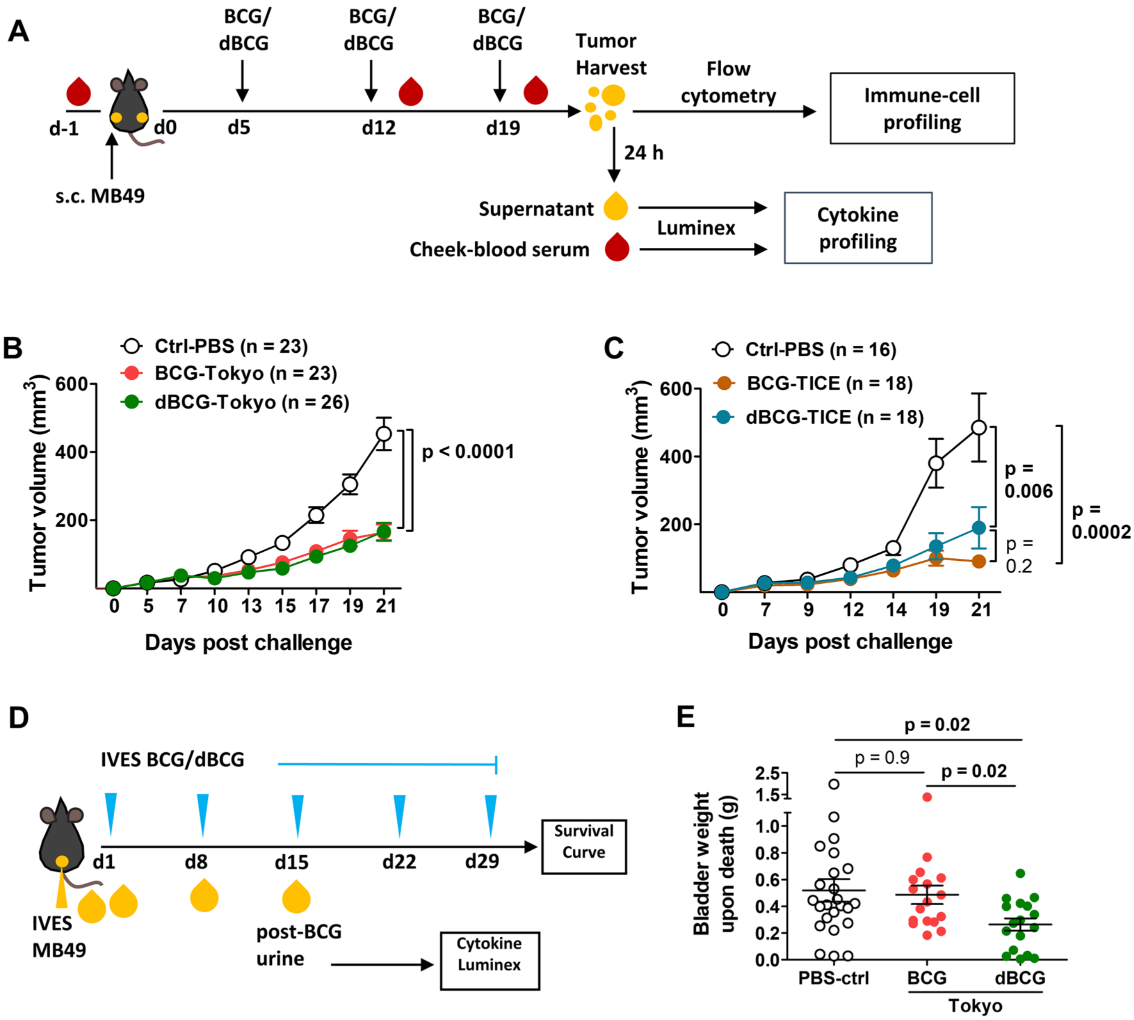


Fig. 1 Selective removal of lipids in BCG does not compromise the treatment efficacy in the mouse bladder tumor model. **a** As indicated in the scheme, WT C57BL/6 mice were subcutaneously challenged with MB49 tumor and treated with either **b** BCG-Tokyo and dBCG-Tokyo or **c** BCG-TICE and dBCG-TICE intratumorally weekly for 3 weeks starting on day 5. Shown are **b** pooled data from 3 independent experiments (n = number of tumors) or **c** pooled data from 2 independent experiments (n = number of tumors). **b**, **c** Two-way ANOVA. Mean ± SEM. **d**, **e** WT female C57BL/6 mice were orthotopically challenged with MB49 tumor and treated with either BCG-Tokyo or dBCG-Tokyo intravesically through bladder instillation via urethral catheter, weekly for 5 weeks starting on day 1. IVES, intravesical (bladder administration). Shown are (**d**) study scheme and (**e**) bladder weight data pooled from 3 independent orthotopic survival experiments (each dot represents one bladder). Mann–Whitney test were applied after normality test. Mean ± SEM

which levels were significantly higher at day 22. In the s.c. tumor, both BCG-Tokyo or dBCG-Tokyo compared to PBS control, induced significant increases of proinflammatory cytokine IFN-γ and IL-1β which favor antitumor response (Fig. 2d, e), whereas other cytokines were not significantly altered (supplemental Fig.S2B), except an increase in IL-10 by dBCG-Tokyo compared to PBS control. Effects on individual serum cytokines were also similar between BCG- and dBCG-TICE groups (not shown), and they both had elevated level of tumor IFN-γ and IL-1β *vs.* control group (Fig. 2f, g). Importantly, the ratio of proinflammatory (i.e., IL-6, TNF-α or IL-1β) to anti-inflammatory (i.e., IL-10) serum cytokines was significantly decreased in dBCG-Tokyo *vs.* conventional BCG-Tokyo (Fig. 2h, i; supplemental Fig.S2C) but with a higher non-significant trend (Fig. 2J) or similar in dBCG-TICE *vs.* BCG-TICE groups (Fig. 2k; supplemental Fig. S2D). These data suggest that delipidation of BCG does not increase systemic or local inflammatory profiles during BCG treatment of mouse bladder cancer and particularly, selective delipidation may reduce inflammatory toxicity induced by BCG-Tokyo strain.

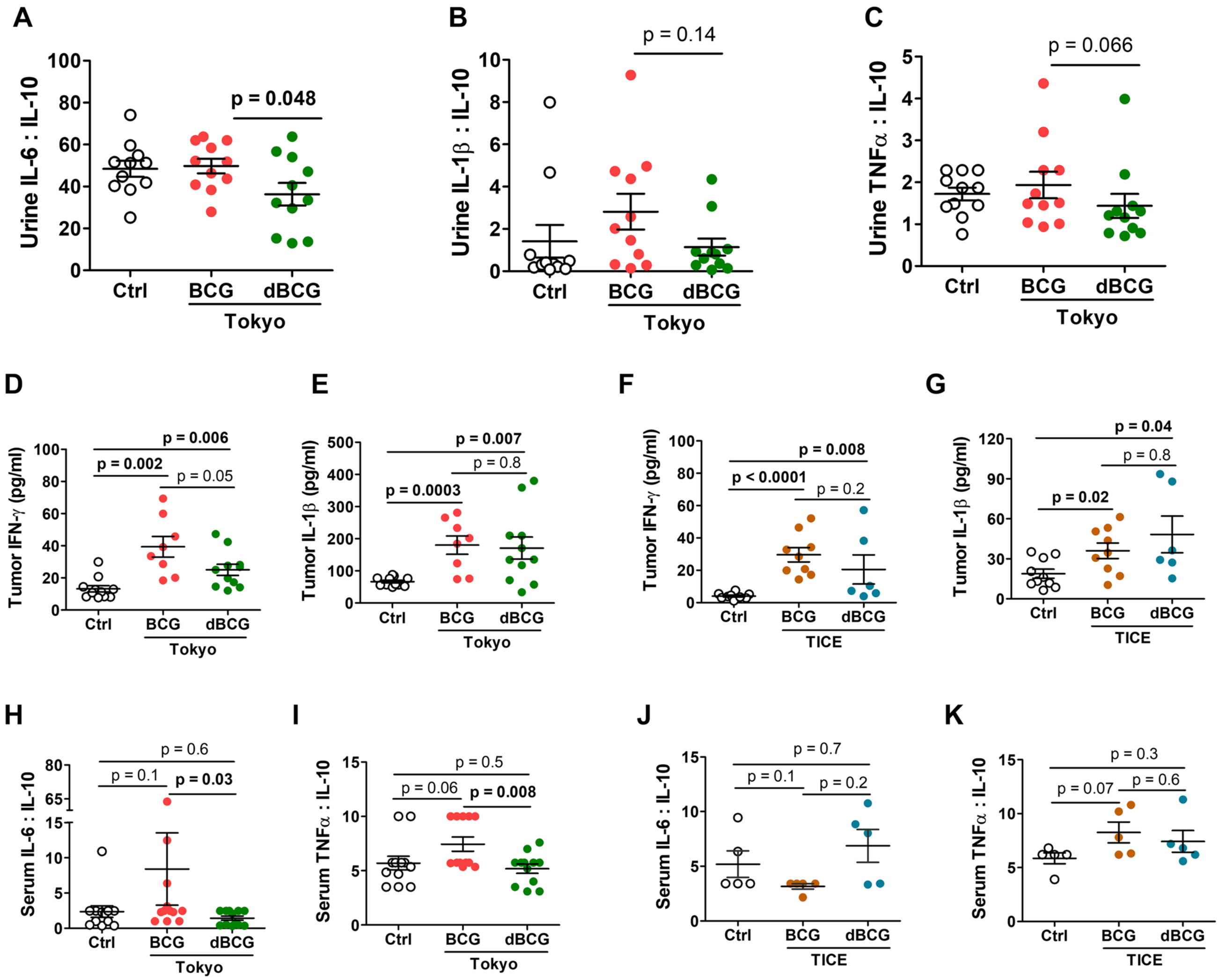


Fig. 2 Selective removal of lipids in BCG-Tokyo can reduce inflammatory toxicity without affecting anti-tumor cytokines. **a–c** Urine collected at various time points (baseline or post-BCG) of orthotopic study were measured for mouse cytokines by Luminex assay (pooled from 2 independent orthotopic experiments; each dot represents one urine sample). Shown are ratios of **a** IL-6 to IL-10, **b** IL-1β to IL-10 by d15 and **c** TNF-α to IL-10 by d1. Student t-test or Mann–Whitney test was applied after normality test. Mean ± SEM. **d–k** At the end of s.c. MB49 tumor study, **d–g** 24-h supernatant from tumor tissue or **h–k** d22 serum were collected for cytokine profiling by Luminex assay. **d–g** one representative experiment (each symbol represents one tumor); **h–i** pooled data from two independent experiments (each symbol represents one mouse); **j–k** one representative experiment (each symbol represents one mouse). **d–k** Student *t*-test or Mann–Whitney test was applied accordingly after normality test. Mean ± SEM

Selective delipidation affects strain-specific BCG's effect on conventional T cells

Conventional αβ T cells contribute to BCG's anti-tumor immunity in bladder cancer [17]. Immune profiling revealed that conventional BCG, including both TICE and Tokyo strains, enhanced tumor-infiltrating $CD8^+$ T cells and their function as expected (Fig. 3a–e). However, selective delipidation reduced BCG-TICE's enhancing effects on $CD8^+$ T cells, particularly for tumor-antigen specific and IFN-γ producing $CD8^+$ T cells (Fig. 3c, d**,** right panels). Conversely, selective delipidation of BCG-Tokyo maintained this strain's boosting effect on tumor-infiltrating $CD8^+$ T cells. Both BCG-Tokyo and dBCG-Tokyo significantly promoted tumor-antigen specific $CD8^+$ T cells as well as IFN-γ and granzyme B producing $CD8^+$ T cells in the tumor (Fig. 3c–e, left panels).

BCG-TICE, however, had more significant effects on promoting tumor-infiltrating $CD4^+$ T cells in contrast to BCG-Tokyo (supplemental Fig.S3A, B–E right vs. left panels). Both BCG-TICE and dBCG-TICE increased activated and degranulating $CD4^+$ T cells in the tumor compared with PBS control, despite less by dBCG-TICE group (supplemental Fig.S3 C–D, right panels). Selective

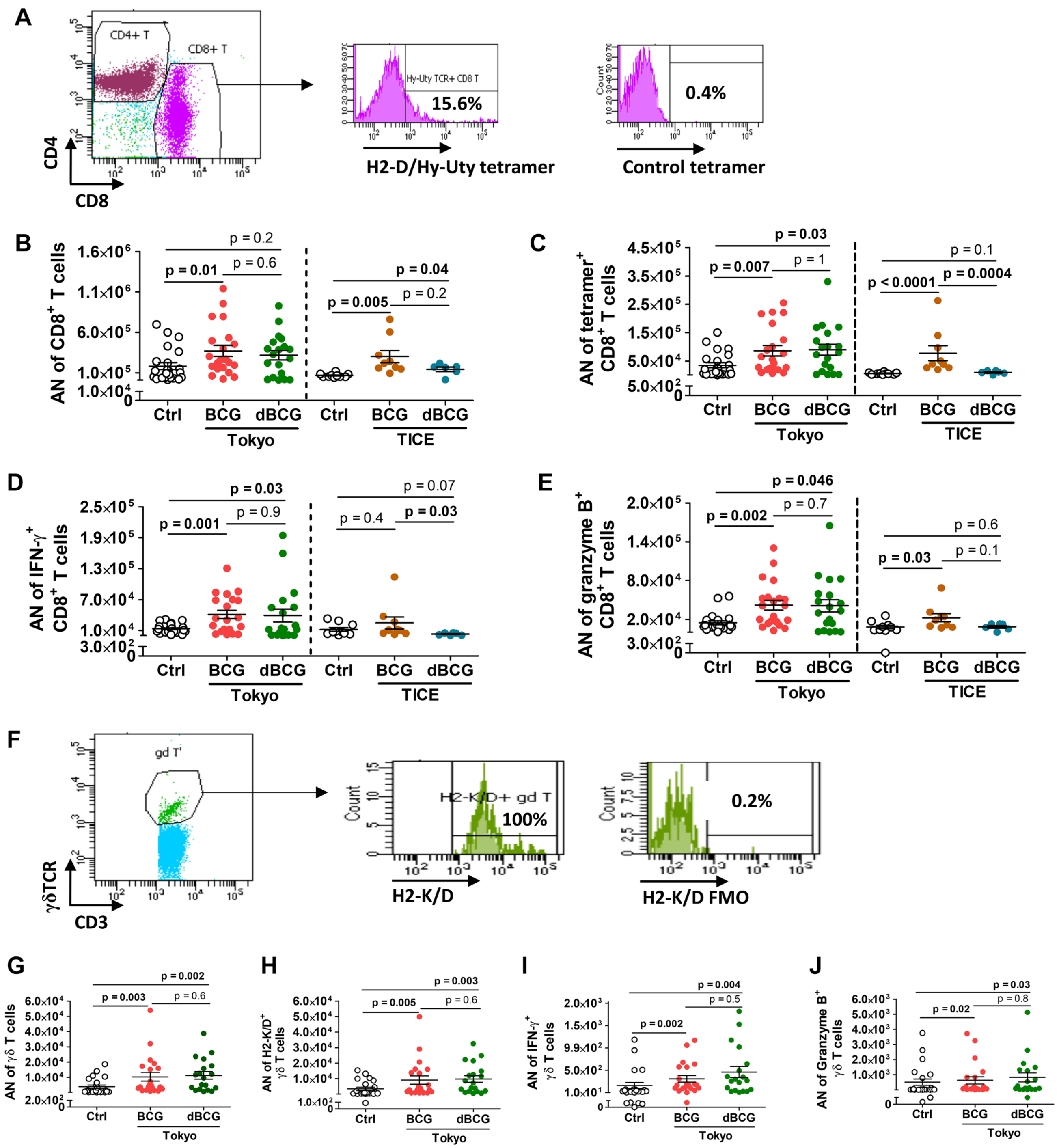


Fig. 3 Delipidation does not compromise the effect of BCG-Tokyo in promoting tumor-infiltrating CD8$^+$ T cells and γδ T cells. s.c. MB49-challenged mice, as described in Fig. 1 legend, were sacrificed on day 22. Tumors were collected and processed for analysis of immune infiltrates by flow cytometry. Shown are: **a** gating strategy and example for tumor-infiltrating CD8$^+$ T cells (gated under live CD45$^+$CD3$^+$ cells) and subsequent markers; **b–e** pooled data from 3 independent experiments of dBCG-Tokyo study (left panels), or one representative experiment of dBCG-TICE study (right panels) for AN of cells per 0.1 g of tumor; **f** gating strategy and example for tumor-infiltrating γδ T cells (gated under live CD45$^+$CD3$^-$ cells) and subsequent markers; **g–j** pooled data from 3 independent experiments of dBCG-Tokyo study for AN of cells per 0.1 g of tumor. One symbol represents one tumor. Student t-test or Mann–Whitney test were applied accordingly after normality test. Mean ± SEM

delipidation decreased BCG-TICE's effect on increasing IL-17 producing $CD4^+$ T cells compared with PBS control (supplemental Fig.S3E, right panel). These data demonstrate that selective delipidation could decrease BCG's effects on promoting antitumor $CD8^+$ and $CD4^+$ T cell in a strain-specific manner. Moreover, BCG-Tokyo may be less sensitive to selective delipidation for $CD8^+$ T cell consequences compared to BCG-TICE.

dBCG retains a robust boost for γδ T or NK cell immune response from different strains

Innate effectors, particularly γδ T cells and NK cells, play important roles in BCG's anti-tumor activity in bladder cancer [16, 18, 19]. Immune profiling from mouse studies shows that BCG-Tokyo and dBCG-Tokyo can both significantly increase the number of tumor-filtrating γδ T cells (Fig. 3f–h). Notably, selective delipidation did not affect BCG-Tokyo's ability to increase γδ TCR and MHC Class I molecules expressed on γδ T cells (supplemental Fig.S4A-B). Selective delipidation also did not alter BCG-Tokyo's impact on activation and degranulation of γδ T cells (supplemental Fig.S4C-D) and their production of IFN-γ and granzyme B (Fig. 3i, j). Although neither BCG-Tokyo nor dBCG-Tokyo increased tumor-infiltrating NK cells compared with control (supplemental Fig.S4E-F), dBCG-Tokyo significantly reduced the surface expression of PD-1 on NK cells and increased their granzyme B production compared with PBS control (supplemental Fig.S4G-H). Thus, selective delipidation did not appear to affect BCG-Tokyo's boosting effect on intratumoral γδ T cells and NK cells.

In contrast, although BCG-TICE promoted activation, degranulation and granzyme B production of tumor-infiltrating γδ T cells compared with control, selective delipidation of BCG-TICE diminished the bacterium's effect on promoting γδ T cells (supplemental Fig.S5A-I). Nonetheless, both BCG-TICE and dBCG-TICE increased tumor-infiltrating and granzyme B-producing NK cells and decreased surface PD-1 expression compared with control (supplemental Fig.S5J-L) which is consistent with other's finding of strong activation of NK cells in vitro by both BCG-TICE and dBCG-TICE [20]. Collectively, these data indicate that impact of BCG delipidation on γδ T cells is strain specific with delipidation of BCG-TICE, but not BCG-Tokyo, resulting in decreased γδ T cells infiltration into the tumor.

dBCG-Tokyo boosts cytotoxic activity of innate immune cells against human bladder cancer

To further investigate effects of selective delipidation on BCG's immune impact, we utilized an in vitro human model.

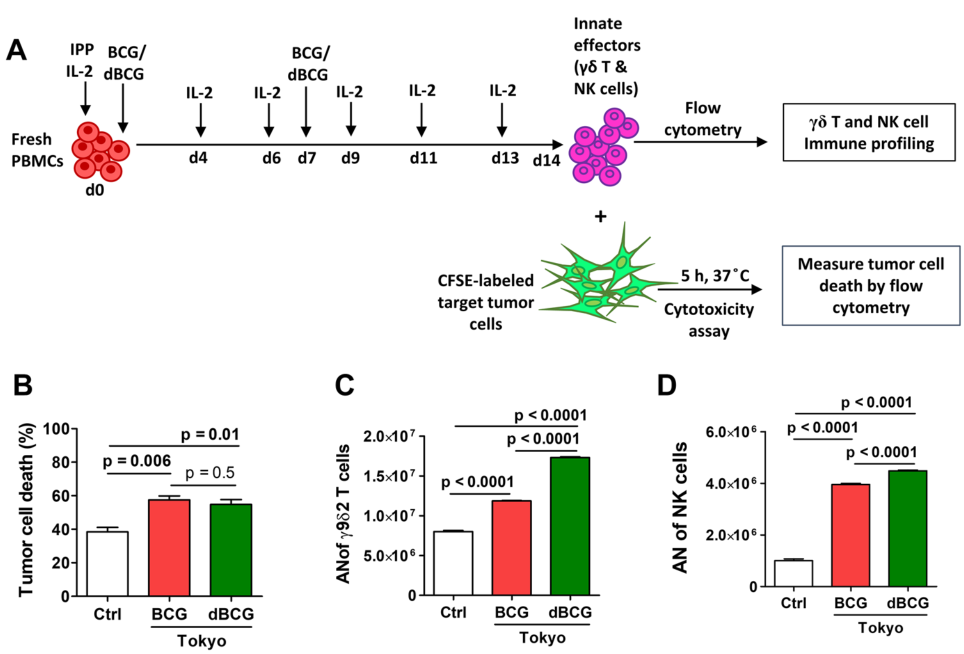


Fig. 4 Delipidated Tokyo strain retains BCG's effect on increasing cytotoxicity and function of expanded human innate effector cells. **a** As indicated in the diagram, PBMC-derived innate effector cells from 3 different bladder cancer patients were expanded for 14 days with or without BCG-Tokyo or dBCG-Tokyo treatment. **b** On day 14, cytotoxicity assays were performed in triplicates against T24 cells at effector to target ratio of 25:1 (pooled results from 3 patients); meanwhile, innate effector cells grew from replicated wells were also analyzed for **c** $CD3^+\gamma9\delta2^+$ T cells and **d** $CD3^-CD56^+$ NK cells by flow cytometry (representative of 3 patients; AN of cells grew from 2×10^6 PBMCs on day 0). Student t-test was applied accordantly after normality test. Mean ± SEM

PBMC-expanded innate effector cells were tested for cytotoxicity against human bladder cancer cell line T24 (Fig. 4a, supplemental Fig.S6A). Both BCG-Tokyo and dBCG-Tokyo increased the cytotoxicity of human innate effector cells compared with control (Fig. 4b) but neither BCG-TICE nor dBCG-TICE did (supplemental Fig.S6B). Immune profiling revealed that selective delipidation does not diminish the promotion of BCG-Tokyo on innate effectors, including γδ T and NK cells (Fig. 4c, d). Furthermore, selective delipidation does not alter the promotion of BCG-Tokyo on NK cells for production of IFN-γ, TNF-α and granzyme B, even though dBCG-Tokyo decreased the production of granzyme B in γδ T compared to BCG-Tokyo (supplemental Fig.S6C-I). Thus, BCG-Tokyo promotes cytotoxicity and function of innate effectors against human bladder cancer cells which is not affected by selective delipidation.

Strain-specific differential profile of lipids on BCG and after selective delipidation

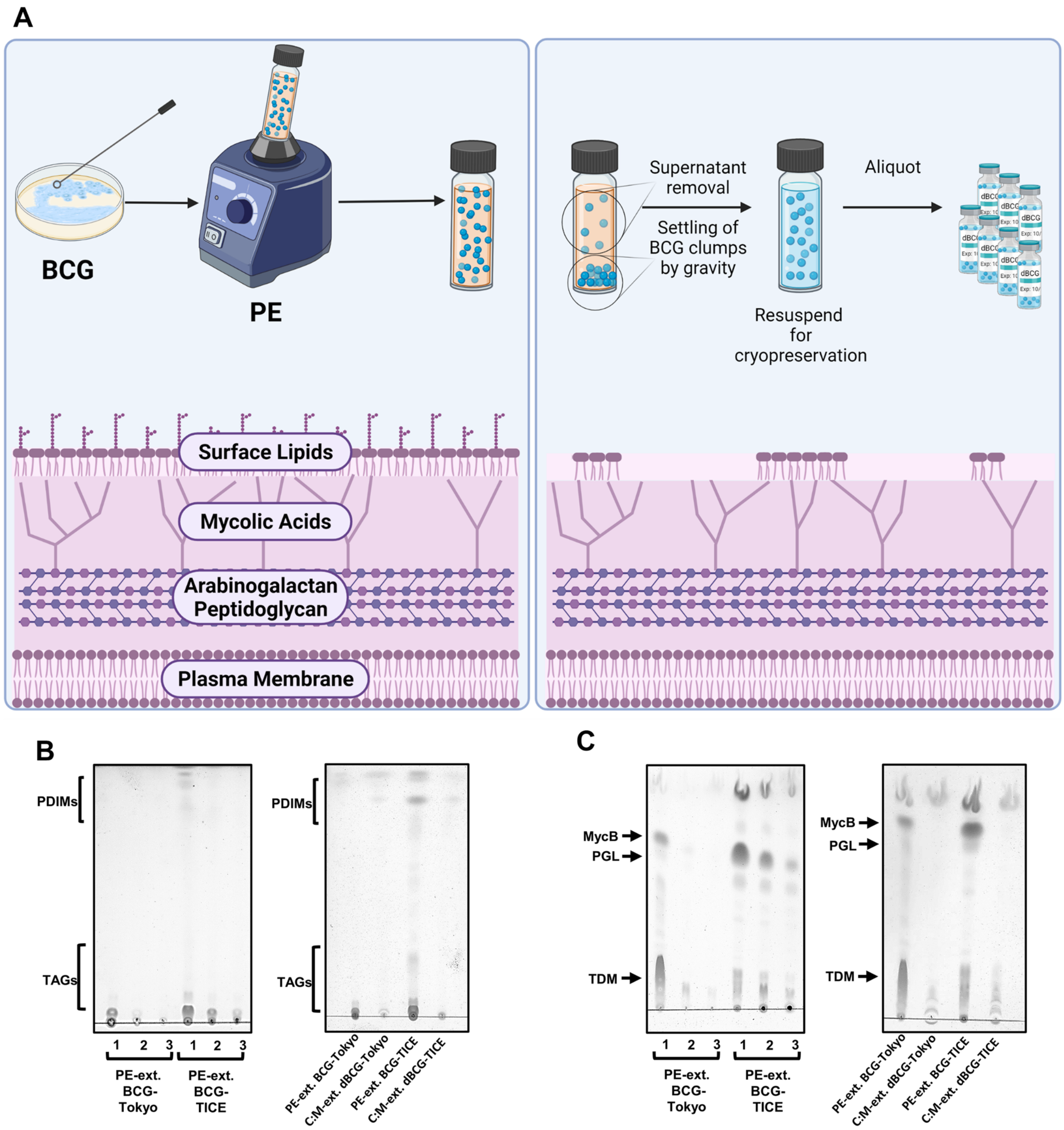
Given the BCG strain-specific immune effects of selective delipidation, we sought to profile lipids in both strains by TLC, as described [7]. PE-extraction removed most apolar lipids from the outer BCG cell envelope (Fig. 5a). Analysis of specific lipids on the BCG cell envelope shows that BCG-Tokyo had less PDIMs, TAGs, MycB and PGL, but more TDM than BCG-TICE (Fig. 5b–d, PE fractions/lanes). In both cases, dBCG-Tokyo and dBCG-TICE retained their viability after being PE-extracted, as observed for BCG-Pasteur [7]. Importantly, after PE-delipidation, although the apolar lipid profile remaining in the cell envelope was similar between dBCG-Tokyo and dBCG-TICE (Fig. 5b, c, C:M fractions/lanes), their polar lipid profile presented striking differences, especially for lower-order PIMs and lipooligosaccharides (LOSs) (supplemental Fig.S7). These data suggest the differential profile of lipids on BCG-Tokyo/dBCG-Tokyo *vs.* BCG-TICE/dBCG-TICE may associate with the difference of their immune effect.

Discussion

Treatment options other than cystectomy for patients with BCG intolerance represent a significant unmet medical need, and many patients suffer from treatment-related adverse events in the course of maintenance therapy [21]. Improved tolerability of BCG could be applied to patients with severe BCG-intolerance and help patients with less severe symptoms complete the full 27 instillation regimen.

Immune toxicity/BCG intolerance is highly associated with bacterial cell-envelope lipids [12], which, however, also potentiate beneficial immune responses needed for effective vaccination and immunotherapy [7, 22]. Complete removal or genetic depletion of cell-envelope lipids could compromise BCG-specific immunity [23], which is important for BCG's efficacy in bladder cancer treatment [16, 24]. Therefore, PE-selective delipidation was adopted to extract virulence-associated apolar lipids without reducing the viability of BCG [7]. Although dBCG might be able to rebuild its cell envelope by restoring lipids in a de novo fashion [25], resynthesized and reconstructed lipids on the BCG cell envelope are unlikely to induce the same amount of toxic inflammatory response as conventional BCG [7], particularly when loss of selective lipids could make the bacterium more susceptible to clearance by the host immune system. Additionally, the majority of BCG should be washed out after post-instillation voiding and any residual bacteria should be excreted in the urine afterwards [26]. For those patients with BCG bacterial persistent in bladder, dBCG could be a good option to delay/prevent the development of toxicity/intolerance.

Further, after PE-delipidation, both dBCG-Tokyo and dBCG-TICE had less apolar lipids in the cell envelope, which decreased their hydrophobicity thereby increasing solubility [27]. Although dBCG-Tokyo retained less PIMs including lower-order PIMs, which stimulate the immune response via toll-like receptor 2 (TLR2) and CD1 [28], it had two species of LOS remaining in its cell envelope when compared to dBCG-TICE, which only had one, despite in higher abundance. Little is known about the role of LOS in inducing immune responses, although it may be an anti-inflammatory suppressor [28]. Of note, PE-delipidation mainly removed large amount of TDM and PGL from the BCG cell envelope, leaving dBCG viable to reconstruct its cell envelope afterwards. We speculate that the lack of these apolar lipids at the initial stages is critical to reduce systemic/local inflammation, but their replenishment in cell envelope is also considered critical, as de novo synthesis of TDM at non-cytotoxic quantities allows dBCG to modulate the local inflammatory response. Since BCG-Tokyo cell envelope contains at least double the amount of TDM compared to BCG-TICE prior to delipidation, dBCG-Tokyo may replenish its cell envelope TDM faster than dBCG-TICE. TDM, a γδ T activator [29], therefore may allow dBCG-Tokyo to retain a stronger effect on promoting γδ T cells in contrast to dBCG-TICE, which may correlate with the finding in this study. Further studies are required to validate the correlation between dBCG-Tokyo *vs.* dBCG-TICE differential lipid profiles and their restoration rate on the cell envelope after delipidation, as well as their immune impact. Additionally, removal of apolar lipids from the BCG cell envelope by PE-delipidation could unmask remaining motifs/ "new" epitopes that may elicit various immune responses. As apolar lipids interact with other cell envelope components, after PE-extraction the exposed motifs can be from polar lipids, proteins, or carbohydrates. These



D

	MycB	PGLs	TDM	PDIMs	TAGs
PE-ext. BCG-Tokyo (*a.u.*)	41,328.04	5,081.57	76,390.11	6,914.32	31,043.21
PE-ext. BCG-TICE (*a.u.*)	106,033.10	6,153.22	33,733.71	53,900.12	96,962.03
PE-ext. BCG-Tokyo/PE-ext.BCG-TICE Ratio	0.389765	0.825839	2.264504	0.12828	0.320158
Log2-Fold Change	**-1.35932**	**-0.27607**	**1.179195**	**-2.96263**	**-1.64314**

Fig. 5 Strain-specific profile of apolar lipids from PE-extracts of BCG-Tokyo and BCG-TICE. **a** Brief scheme (created using Biorender.com) showing apolar lipids being removed from the surface of BCG by sequential 100% petroleum ether (PE) extractions without compromising the cell envelope integrity leaving the dBCG bacteria viable. **b** Apolar lipids PDIMs and TAGs were analyzed by thin layer chromatography (TLC) using a solvent system of PE/acetone (94:4, v/v). Left: Lipids obtained from each sequential PE-extraction (numbers indicate extraction times). Right: Comparison between PE-extracted lipids (PE fraction, pooled from all 3 PE extractions/PE-ext.), and lipids remaining on the cell envelope of dBCG strains (C/M fraction/C:M-ext.). **c** Apolar lipids MycB, PGLs and TDM were analyzed by TLC using a solvent system of C/M (95:5, v/v). Left: Lipids obtained from each sequential PE-extraction (Numbers indicate extraction times). Right: Comparison between PE-extracted lipids (PE fraction, pooled from all 3 PE extractions/PE-ext.), and the lipids remaining on the cell envelope of dBCG strains (C/M fraction/C:M-ext.) **d** Densitometry analysis and Log_2-fold changes of the PE-extracted lipids of BCG-Tokyo compared to those of BCG-TICE. Abbreviations: ext., extracts of; *a.u.*, arbitrary units; PE, petroleum ether; C:M, chloroform/methanol (2:1, v/v, lipid extracts); MycB, mycoside B; PDIMs, phthiocerol dimycocerosates; PGL, phenolic glycolipid; TAGs, triacylglycerols; TDM, trehalose dimycolate

"new" epitopes or motifs may be different between TICE and Tokyo morphologically, structurally, qualitatively and/or quantitatively. Further investigations are required to address any differences of the cell envelope between dBCG strains at the protein, carbohydrate level, as well as the presence or absence of specific lipoglycans that regulate the mycobacterial immune response. These potential differences are being currently studied and will be reported in the future.

As suggested in the study of the mouse model of tuberculosis, dBCG could potentially serve as a safe and efficacious lung vaccine against *M. tuberculosis* infection by promoting a rapid innate immune response [7], which echoes our findings where enhanced innate effector cell function was retained in dBCG-Tokyo strain against bladder cancer cells. Furthermore, as a less toxic immunotherapeutic candidate for NMIBC, selective removal of lipids from the BCG cell envelope could reduce its survival in vivo, which allows the development of either more immunogenic route or more tolerable and frequent treatment regimen using dBCG in bladder cancer patients. The current selective lipid extraction procedure is relatively fast, highly reproducible and inexpensive, an important factor in consideration for implementing into commercial production of dBCG.

Another significance of this study is to meet the need of comparing both TICE- and Tokyo-BCG strains. BCG-Tokyo could be an alternative strain for use in the U.S., but it still induces high toxicity in patients [30–32], which makes the selective delipidation an exciting strategy and set the stage for a phase I clinical testing of dBCG safety and tolerability in patients with bladder cancer.

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Authors' contributions NJ contributed to concept, design, developing in vivo and in vitro models, supervising and performing studies, data acquisition and analysis, manuscript drafting and editing; ML contributed to designing and performing studies, data acquisition and analysis, manuscript editing; AGV contributed to preparation of delipidated BCG and control, protocol optimization, BCG cell-envelope lipids/delipidation analysis; RA contributed to concept and delipidated BCG preparation and protocol development; JIM contributed to concept and delipidated BCG protocol development; KAY contributed to delipidated BCG preparation; NM participated in developing mouse model and draft editing; TJC contributed to resources and expertise for acquiring flow cytometry data; HD contributed to grant acquisition and editing; JBT contributed to key concept, design, delipidated BCG protocol development, manuscript editing, grant acquisition; RSS contributed to key concept, design, grant acquisition, drafting manuscript and editing.

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Declarations

Conflict of interest The studies related to this manuscript were conducted abided by research ethics and approved IACUC (#20120040AR) for animal or clinical sample use (IRB protocol HSC2012-159H). All authors have consent for publishing the data from these studies. Corresponding author R.S.S. discloses other roles as Consultant for FerGene and Clinical Research Support for JBL (SWOG), FKD and Decipher Biosciences. All other authors disclose no conflict of interest.

References

1. Power NE, Izawa J (2016) Comparison of guidelines on non-muscle invasive bladder cancer (EAU, CUA, AUA, NCCN, NICE). Bladder Cancer 2:27–36
2. Sylvester RJ, Van der MA, Lamm DL (2002) Intravesical bacillus Calmette–Guerin reduces the risk of progression in patients with superficial bladder cancer: a meta-analysis of the published results of randomized clinical trials. J Urol 168:1964–1970
3. Bohle A, Jocham D, Bock PR (2003) Intravesical bacillus Calmette–Guerin versus mitomycin C for superficial bladder cancer: a formal meta-analysis of comparative studies on recurrence and toxicity. J Urol 169:90–95
4. van der Meijden AP et al (2001) Intravesical instillation of epirubicin, bacillus Calmette-Guerin and bacillus Calmette–Guerin plus isoniazid for intermediate and high risk Ta, T1 papillary carcinoma of the bladder: a European Organization for Research and Treatment of Cancer genito-urinary group randomized phase III trial. J Urol 166:476–481
5. Babjuk M et al (2011) EAU guidelines on non-muscle-invasive urothelial carcinoma of the bladder, the 2011 update. Eur Urol 59:997–1008
6. Zlotta AR, Fleshner NE, Jewett MA (2009) The management of BCG failure in non-muscle-invasive bladder cancer: an update. Can Urol Assoc J 3:S199-205
7. Moliva JI et al (2019) Selective delipidation of Mycobacterium bovis BCG enables direct pulmonary vaccination and enhances protection against Mycobacterium tuberculosis. Mucosal Immunol 12:805–815
8. Geisel RE, Sakamoto K, Russell DG, Rhoades ER (2005) In vivo activity of released cell wall lipids of Mycobacterium bovis bacillus Calmette–Guerin is due principally to trehalose mycolates. J Immunol 174:5007–5015
9. Chauhan P et al (2013) Secretory phosphatases deficient mutant of Mycobacterium tuberculosis imparts protection at the primary site of infection in guinea pigs. PLoS ONE 8:e77930
10. Astarie-Dequeker C et al (2009) Phthiocerol dimycocerosates of *M. tuberculosis* participate in macrophage invasion by inducing changes in the organization of plasma membrane lipids. PLoS Pathog 5:e1000289
11. Cambier CJ, O'Leary SM, O'Sullivan MP, Keane J, Ramakrishnan L (2017) Phenolic glycolipid facilitates mycobacterial

escape from microbicidal tissue-resident macrophages. Immunity 47:552-565e554
12. Reed MB et al (2004) A glycolipid of hypervirulent tuberculosis strains that inhibits the innate immune response. Nature 431:84–87
13. Day TA et al (2014) Mycobacterium tuberculosis strains lacking surface lipid phthiocerol dimycocerosate are susceptible to killing by an early innate host response. Infect Immun 82:5214–5222
14. Arbues A et al (2016) Trisaccharides of phenolic glycolipids confer advantages to pathogenic mycobacteria through manipulation of host-cell pattern-recognition receptors. ACS Chem Biol 11:2865–2875
15. Ji N et al (2021) Gammadelta T cells support antigen-specific alphabeta T cell-mediated antitumor responses during BCG treatment for bladder cancer. Cancer Immunol Res 9:1491–1503
16. Ji N et al (2019) Percutaneous BCG enhances innate effector antitumor cytotoxicity during treatment of bladder cancer: a translational clinical trial. Oncoimmunology 8:1614857
17. Ratliff TL, Ritchey JK, Yuan JJ, Andriole GL, Catalona WJ (1993) T-cell subsets required for intravesical BCG immunotherapy for bladder cancer. J Urol 150:1018–1023
18. Ji N et al (2021) Rapamycin enhances BCG-specific gammadelta T cells during intravesical BCG therapy for non-muscle invasive bladder cancer: a randomized, double-blind study. J Immunother Cancer 9:e001941
19. Mukherjee N et al (2018) Intratumoral CD56(bright) natural killer cells are associated with improved survival in bladder cancer. Oncotarget 9:36492–36502
20. Esteso G et al (2021) Natural killer anti-tumor activity can be achieved by in vitro incubation with heat-killed BCG. Front Immunol 12:622995
21. Lamm DL et al (2000) Maintenance bacillus Calmette–Guerin immunotherapy for recurrent TA, T1 and carcinoma in situ transitional cell carcinoma of the bladder: a randomized Southwest oncology group study. J Urol 163:1124–1129
22. Tran V, Ahn SK, Ng M, Li M, Liu J (2016) Loss of lipid virulence factors reduces the efficacy of the BCG vaccine. Sci Rep 6:29076
23. Silva CL, Ekizlerian SM, Fazioli RA (1985) Role of cord factor in the modulation of infection caused by mycobacteria. Am J Pathol 118:238–247
24. Biot C et al (2012) Preexisting BCG-specific T cells improve intravesical immunotherapy for bladder cancer. Sci Transl Med 4:137ra172
25. Kawai Y, Mercier R, Errington J (2014) Bacterial cell morphogenesis does not require a preexisting template structure. Curr Biol 24:863–867
26. Durek C et al (2001) The fate of bacillus Calmette–Guerin after intravesical instillation. J Urol 165:1765–1768
27. Jankute M et al (2017) The role of hydrophobicity in tuberculosis evolution and pathogenicity. Sci Rep 7:1315
28. Garcia-Vilanova A, Chan J, Torrelles JB (2019) Underestimated manipulative roles of mycobacterium tuberculosis cell envelope glycolipids during infection. Front Immunol 10:2909
29. Hoq MM et al (1997) Role of gamma delta TCR+ lymphocytes in the augmented resistance of trehalose 6,6'-dimycolate-treated mice to influenza virus infection. J Gen Virol 78(Pt 7):1597–1603
30. Boehm BE et al (2017) Efficacy of bacillus Calmette–Guerin strains for treatment of nonmuscle invasive bladder cancer: a systematic review and network meta-analysis. J Urol 198:503–510
31. Miyazaki J, Onozawa M, Takaoka E, Yano I (2018) Bacillus Calmette–Guerin strain differences as the basis for immunotherapies against bladder cancer. Int J Urol 25:405–413
32. Koguchi D et al (2020) Impact of maintenance therapy using a half dose of the bacillus Calmette-Guerin Tokyo strain on recurrence of intermediate and high-risk nonmuscle invasive bladder cancer: a retrospective single-center study. BMC Urol 20:194